

LUYAO NIU (SHE/HER)

PhD student, University of Southern California
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EDUCATION

University of Southern California

PhD in Transportation Systems Engineering | Aug. 2026 - Present

Peking University

M.Sc. in Urban Planning (Smart City and Big Data Program) | Sept. 2023 - Jun. 2026 | GPA: 3.75/4

Beijing Jiaotong University

B.Sc. in Traffic Engineering; Minor in Finance | Sept. 2019 - Jun. 2023 | GPA: 3.97/4 | Top 0.5% | Beijing Outstanding Graduate

RESEARCH INTERESTS

Learning, decision-making, and coordination in intelligent systems, with a growing focus on multi-agent and autonomous systems.

Learning & Reasoning | Multi-Agent Decision-Making | Coordination & Collective Intelligence

Current topic at USC: Coalition coordination in mixed-autonomy traffic

When should autonomous-vehicle firms route independently, coordinate in partial coalitions, or share a common dispatcher? I study how these organizational choices affect traffic equilibrium and network performance when human drivers also adapt their routes.

SELECTED RESEARCH EXPERIENCE

Joint Design of Shared Autonomous Vehicles and Public Transit Systems for Tourist Cities

- Formulated an integrated Shared Autonomous Vehicles (SAV)-transit design problem accommodating heterogeneous demand types (normal trips with carpooling and private tour trips requiring privacy), balancing system efficiency and accessibility in tourist city contexts.
- Applied continuum approximation and network queuing theory to derive flow equilibrium equations; formulated mixed integer program minimizing total system cost (agency and user costs) to determine optimal SAV fleet size and FRT network layout.
- Achieved 15-25% total cost reduction in low-demand scenarios compared to transit-only baselines; analyzed sensitivity of design outcomes to private tour trip proportions and demand levels; paper published in Transportation Research Part C .

Optimized Path Planning and Traffic Flow Assignment Model Balancing User Equilibrium and System Optimum

- Developed a traffic assignment framework balancing user equilibrium and system optimum to mitigate congestion caused by homogeneous navigation recommendations; conducted SP survey and used discrete model to elicit heterogeneous preferences and constructed personalized alternative route sets for systemoptimal assignment.
- Validated framework through simulation experiments on Beijing road network, demonstrating reductions in total travel time and congestion index; won First Prize in National University Traffic Technology Competition.
- Currently advancing research by applying inverse reinforcement learning on mobile signaling and trajectory data to better identify individual preferences and travel intentions; extending framework to connected and autonomous vehicle scenarios with centralized routing control.

Event-CausNet: Unlocking Causal Knowledge from Text with Large Language Models for Reliable Spatio-Temporal Forecasting

- Addressed limitations in spatio-temporal GNNs during non-recurring traffic events (accidents, work zones) by developing a causal inference framework that uses LLMs to extract and quantify causal knowledge from unstructured event reports, building a Causal Knowledge Base through Average Treatment Effects estimation.
- Engineered a dual-stream architecture integrating the Causal Knowledge Base with GNN-LSTM networks via novel Causal Attention mechanism to adjust traffic forecasts based on event-driven causal factors.
- Led full research execution from problem formulation to experimental validation; evaluated the framework on real-world datasets; submitted to KDD 2027.

Behavioral Changes from Electric Vehicles Adoption: A Quasi-experimental Study

- Built large-scale quasi-experimental study using mobile signaling data to estimate causal impacts of EV adoption on travel behavior, overcoming limitations of small-sample survey methods in existing literature.
- Developed novel identification strategy leveraging APP usage logs (charging and brand-specific apps) to identify EV adopters as treatment group and construct matched control groups for Difference-in-Differences causal inference analysis.
- Revealed car usage rebound effects (strongest during off-peak and non-commute periods), increased active travel linked to charging infrastructure access, and metro substitution effects;

AskNearby: An LLM-based Application for Neighborhood Information Retrieval and Personalized Cognitive-map Recommendations

- Identified the Local Life Information Accessibility (LLIA) gap in 15-minute city initiatives, where existing location-based systems neglect spatial, temporal, and cognitive factors critical for hyperlocal decisionmaking and community engagement.
- Developed AskNearby, an LLM-powered community platform integrating a three-layer Retrieval-Augmented Generation (RAG) pipeline (combining graph-based, semantic-vector, and geographic retrieval) with a cognitive-map model encoding users' neighborhood familiarity and preferences for context-aware recommendations.
- Achieved significant improvements over LLM-based and map-based baselines in retrieval accuracy and recommendation quality through real-world deployments; demonstrated robust spatiotemporal grounding and cognitive-aware ranking capabilities; paper accepted for presentation by ACM SIGSPATIAL GeoAI'25.

Multi-Agent Coordination and Collision Avoidance for Unmanned Warehouse Systems

- Proposed a hierarchical scheduling framework modeling multi-AGV task allocation as a Colored Traveling Salesman Problem; integrated Mixed-Integer Linear Programming for global order optimization with A* algorithm for precise pathfinding.
- Designed a dynamic zoning and load-balancing mechanism based on Monte Carlo Tree Search and UCT, optimizing pallet-to-workstation assignment to alleviate local congestion and minimize empty-load travel distance.
- Constructed a deadlock-free collision avoidance policy using Deep Reinforcement Learning with CNNbased state representation, ensuring robust navigation in dense environments; won Second Prize in China MathorCup Modeling Competition.

WORK EXPERIENCE

AI Agent Engineering Intern | May - Aug. 2026

Xiaohongshu · Data Analytics

- Domain knowledge infrastructure: Built a domain-specific data analytics knowledge base, structuring analytical knowledge to support knowledge-grounded agent workflows.
- Agent architecture and development: Developed the agent framework and implemented data analysis workflows, integrating knowledge resources and domain capabilities into the agent architecture.
- End-to-end validation: Validated the complete agent workflow, evaluating execution behavior and analytical outputs to guide systematic iteration.
- Domain Skill lifecycle: Designed and built domain-specific Skills, covering evaluation, monitoring, and ongoing operations to support maintainable, reusable agent capabilities.

Research Assistant | Jan. 2025 - Present

Data Science Department, Duke Kunshan University | Suzhou, China

- Led and independently executed research projects focusing on spatiotemporal data mining, model development, and traffic flow forecasting.
- Independently completed two studies: one published in IEEE GLOBECOM and another submitted to KDD 2027 (Event-CausNet).

EXTRA-CURRICULAR ACTIVITIES

President, Student Union | Jun. 2021 - Jun. 2023

Beijing Jiaotong University | Beijing, China

Volunteer, Beijing 2022 Winter Olympics | Jan. 2022 - Apr. 2022

National Indoor Stadium | Beijing, China

SELECTED AWARDS & HONORS

National Scholarship (Highest undergraduate honor in China) Dec. 2020

First-class Academic Scholarship (Top 1%) Dec. 2020

Baosteel Outstanding Student Award (Only 500 recipients nationwide annually) Dec. 2022

First Prize, China University Traffic & Transportation Technology Competition Jul. 2022

Second Prize, China MathorCup Optimizing Mathematical Modeling Competition Apr. 2022

Honorable Mention, Mathematical Contest in Modeling (MCM/ICM) Apr. 2022

Third Prize, China College Mathematics Competition Apr. 2020

Third Prize, Baidu ERNIE Cup AI Entrepreneurship Competition Sep. 2024

SKILLS

Languages: Mandarin Chinese (Native); English (Fluent, TOEFL iBT 101); Spanish (Conversational)

Programming & Data: Python (Pytorch, Scikit-learn, Geopandas, Networkx), C, MATLAB, R, SQL

Research Methods: Optimization, Machine Learning, Deep Learning, Reinforcement Learning, LLMs

PUBLICATIONS

Journal

Shi, Z., Luo, S., **Niu, L.**, Shao, C. (2024). Joint Design of Public Transit and Autonomous Vehicle Mobility Systems for Tourist Cities Considering Abnormal Trips. *Transportation Research Part C*. doi: 10.1016/j.trc.2025.105179.

Niu, L., Wang, Y., He, S., Zhang, W.(2025). A Cross-Lingual Framework driven by Large Language Models for Spatial Identity Sensing from Social Media. (Under review)

Liu, C., **Niu, L.**, Wan, L., Zhang, W. (2025) Travel Behavior Shifts Following Electric Vehicle Adoption: Evidence from a Big Data-driven Quasi-experiment. (Research manuscript)

Zhang, W., Wang, Y., Huang, N., **Niu, L.**, Wu, Y., Niu, C. (2025) Large-scale longitudinal mobility data reveal a causal view of social influence's amplification after risk. (Submitted to *Nature Human Behavior*)

Niu, L., Xing, X., Liu, Z., Yao, E. (2025) Traffic assignment Model balancing User Preference and system optimum. (In preparation)

Conferences

Niu, L., Wang, Z., Liu., J., Boukerche, A., Sun, P. (2025). MF-AttnBiLSTM: Traffic Flow Prediction via Hybrid Signal Decomposition and Dual-Stream Temporal Attention Learning. *IEEE GLOBECOM 2025*, Oral Presentation.

Niu, L., Deng, Z., Li, B., Huang, N., Liu, R., Zhang, W.(2025). AskNearby: An LLM-based Application for Neighborhood Information Retrieval and Personalized Cognitive-map Recommendations. *ACM SIGSPATIAL GeoAI'25*, Oral Presentation. doi: 10.48550/arXiv.2512.02502

Niu, L., Wang, Y., He, S., Zhang, W. (2025). Evaluating the Causal Impacts on Spatial Satisfaction Sentiment of Shenzhen-Hong Kong Regional Integration Policies: An Interrupted Time Series Study Leveraging Large Language Models. *EUGEO 2025 Proceedings*, Oral Presentation.

Niu, L., Gu, J., Liu, C., Zhang, W. (2025). Why Public Transport Policies Work or Not? A Metaanalysis Based on the Behaviour Change Wheel. *AESOP 2025 Proceedings*, Oral Presentation. (In preparation for submission to *Transport Reviews*)

Niu, L., Wang, Z., Guan, S., Sun, P. (2025) Event-CausNet: Unlocking Causal Knowledge from Text with LLMs for Reliable Spatio-Temporal Forecasting. (Submitted to *KDD 2027*) doi: 10.48550/arXiv.2511.12769

Niu, L., Huang, N.(2025) ST-ProC: A Graph-Prototypical Framework for Robust Semi-Supervised Travel Mode Identification. (Preprint) doi: 10.48550/arXiv.2511.13702

Zhu, H., Zhang, W., Huang, N., Li, B., **Niu, L.**,...,Liu, X. (2024). PlanGPT: Enhancing urban planning with tailored language model and efficient retrieval. (Preprint) doi: 10.48550/arXiv:2402.19273